

claude-windows / 42211f80-2169-4025-ad9d-99c1aa29ff70

Metadata

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- Kind: `claude`  
- Source file: `C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\42211f80-2169-4025-ad9d-99c1aa29ff70\subagents\workflows\wf_6237b110-d58\agent-a47b9b8be6196f9b2.jsonl`  
- SHA-256: `1627b304f40056755bec026d8b3f2ec04f2a8d5a7cc8fcf94f07df42d9aaa2d1`  
- Source modified: `2026-06-06T17:34:49+00:00`  
- Imported at: `2026-07-05T16:48:28+00:00`  
- project: `wf_6237b110-d58`  
- session_id: `42211f80-2169-4025-ad9d-99c1aa29ff70`
```

Transcript

1. user (2026-06-06T17:32:36.251Z)

Adversarial Claim Verifier (voter 1/3)

Be SKEPTICAL. Try to REFUTE this claim. ?2/3 refutations kill it.

Research question

Research alternative methods to automatically detect rally start/end segments (temporal segmentation) in racket-sport video ? especially badminton ? for a SELF-HOSTED, LOCAL-FIRST pipeline, and assess whether a cloud-VLM teacher ? local-model distillation is the best approach or whether other methods are competitive or complementary. Keep it focused ("mini" research): prioritize actionable, commercially-licensable, locally-runnable approaches over exotic SOTA needing huge compute or restrictive licenses.

OUR CONTEXT / CONSTRAINTS (ground every recommendation in these):

- Self-hosted on a single consumer GPU (RTX 2070 Super), Windows + WSL2. We want LOCAL-ONLY inference at runtime ? privacy lever: user video never leaves the device. It's a commercial product, so PERMISSIVE / commercial-friendly licenses are REQUIRED (avoid AGPL and non-commercial-only weights/datasets).
- Current substrate: WASB (MIT) shuttle-trajectory detector ? per-frame (x, y, visibility). We derive rally windows via velocity/merge-gap/min-duration thresholds plus a net-crossing gate. We ALSO have a permissive person detector available (torchvision Faster R-CNN + supervision ByteTrack) and the video's AUDIO track.
- Current best approach: use paid Gemini (cloud VLM, full-context) OFFLINE as a TEACHER to label rallies, then DISTILL into a local model (logistic regression on shuttle-trajectory features). Findings so far: threshold-tuning ceilings around F1 ? 0.44 and overfits with few videos; the learned model is data-starved at n=2 labeled videos; WASB candidate RECALL is a real bottleneck (its windows sometimes don't align with true rallies at IoU 0.5).
- Inputs we can exploit locally: video frames, the shuttle trajectory, player bounding boxes / pose (if we add a permissive pose model), and the AUDIO track (shuttle "thwack" hits, crowd noise, umpire/score calls).

SPECIFICALLY COVER:

1. Method families for rally/point segmentation in racket sports (badminton, tennis, table tennis, pickleball): audio-based shot/hit detection; temporal action localization (TAL) deep models; pose/player-stance + court-homography state machines (e.g., "serve-ready stance" gating); classical change-point detection / HMM on the shuttle trajectory; scoreboard / score-overlay OCR for point boundaries; self-supervised / weakly-supervised approaches; and VLM teacher?student distillation.
2. For EACH family: how it works, the labels/data it needs, expected accuracy, local-inference COMPUTE cost, and LICENSE / commercial viability of any models, datasets, or weights involved.
3. Badminton rally-segmentation datasets (e.g., ShuttleSet, TrackNet/TrackNetV2 data) and their licenses / commercial usability.
4. A RANKED recommendation for our local-first, permissive, single-GPU setup: is Gemini-teacher ? local-distillation the best, or are some alternatives competitive or complementary ?

especially AUDIO-based hit detection and pose/court-geometry state machines, which we can run fully locally?

5. Concrete, cheap next experiments we could run with what we already have (shuttle trajectory + person detector + audio).

Claim under review

"Badminton point/rally segmentation can be done by a per-frame CNN image classifier that distinguishes broadcast-camera rally frames from non-rally frames (replays/close-ups), achieving F1 97.23 / Precision 98.06 / Recall 92.34 ? without any shuttle tracking."

****Source:**** <https://arxiv.org/pdf/2308.12199> (primary)

****Supporting quote:**** "Rally frames are usually always shot from the broadcast camera view whereas the non-rally frames can include close-ups. ... CNN: F1-Score 97.23, Precision 98.06, Recall 92.34"

Checklist

1. Is the claim actually supported by the quote, or is it an overreach/misread?
2. WebSearch for contradicting evidence ? does any credible source dispute or heavily qualify this?
3. Is the source quality sufficient for the claim's strength? (extraordinary claims need primary sources)
4. Is the claim outdated? (check dates ? old claims about fast-moving fields are suspect)
5. Is this a marketing claim / press release / cherry-picked benchmark / forum speculation?

****refuted=true**** if: unsupported by quote / contradicted / low-quality source for strong claim / outdated / marketing fluff.

****refuted=false**** ONLY if: claim is well-supported, current, and source quality matches claim strength.

Default to refuted=true if uncertain.

Structured output only. Evidence MUST be specific.

2. assistant (2026-06-06T17:32:39.237Z)

I'll verify this claim about per-frame CNN image classification for rally segmentation. Let me start by examining the primary source.

3. assistant (2026-06-06T17:32:43.270Z)

I'll fetch the primary source to verify the exact claim, numbers, and methodology.

4. user (2026-06-06T17:32:46.949Z)

maxContentLength size of 10485760 exceeded

5. assistant (2026-06-06T17:32:49.551Z)

The PDF is too large. Let me find the abstract page and search for the paper's content.

6. user (2026-06-06T17:32:57.475Z)

Paper Summary

****Title:**** Towards Real-Time Analysis of Broadcast Badminton Videos

****Main Topic:**** The paper addresses player movement analysis in badminton using live broadcast video feeds with computer vision techniques.

****Abstract (Verbatim):****

"Analysis of player movements is a crucial subset of sports analysis. Existing player movement analysis methods use recorded videos after the match is over. In this work, we propose an end-to-end framework for player movement analysis for badminton matches on live broadcast match

videos. We only use the visual inputs from the match and, unlike other approaches which use multi-modal sensor data, our approach uses only visual cues. We propose a method to calculate the on-court distance covered by both the players from the video feed of a live broadcast badminton match. To perform this analysis, we focus on the gameplay by removing replays and other redundant parts of the broadcast match. We then perform player tracking to identify and track the movements of both players in each frame. Finally, we calculate the distance covered by each player and the average speed with which they move on the court. We further show a heatmap of the areas covered by the player on the court which is useful for analyzing the gameplay of the player. Our proposed framework was successfully used to analyze live broadcast matches in real-time during the Premier Badminton League 2019 (PBL 2019), with commentators and broadcasters appreciating the utility."

****Regarding Rally Classification:**** The abstract does not discuss rally vs non-rally frame classification using CNNs or rally segmentation methodology.

7. user (2026-06-06T17:32:59.712Z)

Web search results for query: "arxiv 2308.12199 badminton rally segmentation CNN frame classification"

Links: [{"title": "Towards Real-Time Analysis of Broadcast Badminton Videos", "url": "https://arxiv.org/pdf/2308.12199"}, {"title": "A Comparative Analysis of Deep Learning Models and Gradient Computation for Rally Detection in Badminton Videos | SN Computer Science | Springer Nature Link", "url": "https://link.springer.com/article/10.1007/s42979-025-03935-0"}, {"title": "A deep learning based framework for badminton rally outcome prediction | IEEE Conference Publication | IEEE Xplore", "url": "https://ieeexplore.ieee.org/iel7/10082768/10082763/10082764.pdf"}, {"title": "MonoTrack: Shuttle trajectory reconstruction from monocular badminton video", "url": "https://arxiv.org/pdf/2204.01899"}, {"title": "A Deep Learning Based Framework for Badminton Rally Outcome Prediction", "url": "https://easychair.org/publications/preprint/1pDZ"}, {"title": "Towards Structured Analysis of Broadcast Badminton Videos", "url": "https://arxiv.org/pdf/1712.08714"}, {"title": "Automated Hit-frame Detection for Badminton Match Analysis", "url": "https://arxiv.org/pdf/2307.16000"}, {"title": "BST: Badminton Stroke-type Transformer for Skeleton-based Action Recognition in Racket Sports", "url": "https://arxiv.org/html/2502.21085v2"}, {"title": "CoachAI: A Project for Microscopic Badminton Match Data Collection and Tactical Analysis", "url": "https://arxiv.org/pdf/1907.12888"}, {"title": "[2308.12199] Towards Real-Time Analysis of Broadcast Badminton Videos", "url": "https://arxiv.org/abs/2308.12199"}]

Based on the search results, I found the paper you're looking for: "Towards Real-Time Analysis of Broadcast Badminton Videos" (arXiv 2308.12199).

Here's what the paper is about:

The work proposes an end-to-end framework for player movement analysis for badminton matches on live broadcast match videos. The approach uses only visual inputs from the match and visual cues rather than multi-modal sensor data.

Regarding rally segmentation specifically, the paper addresses this through what it calls "Point Segmentation." A "rally" frame is defined as a frame where the actual gameplay is happening and a "non-rally" frame is a frame where the replay (highlights) has been shown, or players are resting, etc. The researchers annotated 23,156 frames in total, where 15,147 were non-rally frames and 8,009 were rally frames.

The paper proposes a method to calculate the on-court distance covered by both players from the video feed, performs player tracking to identify and track the movements of both players in each frame, and finally calculates the distance covered by each player and the average speed with which they move on the court.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

8. assistant (2026-06-06T17:33:03.174Z)

The paper is confirmed as "Towards Real-Time Analysis of Broadcast Badminton Videos." The search results confirm a "Point Segmentation" module with rally vs non-rally frame classification and the dataset of 23,156 frames. Let me verify the exact CNN metrics and how the paper characterizes the task, plus check the HTML version and any independent assessment.

9. user (2026-06-06T17:33:08.505Z)

I cannot provide a meaningful answer to your question based on the provided content. The document appears to be a corrupted or improperly encoded PDF file containing primarily binary image data and stream information, making it impossible to extract coherent text about badminton video analysis, frame classification methods, or CNN-based segmentation techniques.

To properly address your inquiry about rally vs. non-rally frame classification or gameplay/replay separation methods, I would need access to a readable academic paper on this topic.

[Binary content (application/pdf, 4.7MB) also saved to C:\Users\avidu\claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\42211f80-2169-4025-ad9d-99c1aa29ff70\tool-results\webfetch-1780767185534-0drige.pdf]

10. user (2026-06-06T17:33:12.468Z)

Web search results for query: "'Towards Real-Time Analysis of Broadcast Badminton Videos' point segmentation CNN F1 97.23 precision 98.06 recall 92.34"

Links: [{"title": "(PDF) Towards Structured Analysis of Broadcast Badminton Videos", "url": "https://www.researchgate.net/publication/322075902_Towards_Structured_Analysis_of_Broadcast_Badminton_Videos"}, {"title": "Towards Structured Analysis of Broadcast Badminton Videos | Request PDF", "url": "https://www.researchgate.net/publication/325002967_Towards_Structured_Analysis_of_Broadcast_Badminton_Videos"}, {"title": "(PDF) Towards Structured Analysis of Broadcast Badminton Videos", "url": "https://www.academia.edu/70243179/Towards_Structured_Analysis_of_Broadcast_Badminton_Videos"}, {"title": "Towards Real-Time Analysis of Broadcast Badminton Videos", "url": "https://arxiv.org/pdf/2308.12199"}, {"title": "[2308.12199] Towards Real-Time Analysis of Broadcast Badminton Videos", "url": "https://arxiv.org/abs/2308.12199"}, {"title": "Towards Structured Analysis of Broadcast Badminton Videos", "url": "https://arxiv.org/pdf/1712.08714"}, {"title": "[PDF] Towards Structured Analysis of Broadcast Badminton Videos | Semantic Scholar", "url": "https://www.semanticscholar.org/paper/Towards-Structured-Analysis-of-Broadcast-Badminton-Ghosh-Singh/a2359c0f81a7eb032cff1fe45e3b80007facaa2a"}, {"title": "Towards Structured Analysis of Broadcast Badminton Videos", "url": "https://cvit.iiit.ac.in/research/projects/cvit-projects/badminton-analytics"}]

I found the paper "Towards Real-Time Analysis of Broadcast Badminton Videos" in the search results. However, the specific performance metrics you mentioned (F1 97.23, precision 98.06, recall 92.34) are not appearing in the search results I retrieved.

The search results show information about a related paper titled "Towards Structured Analysis of Broadcast Badminton Videos" which reports different metrics: 95.44% point segmentation accuracy, 97.38% player detection score (mAP@0.5), 97.98% player identification accuracy, and stroke segmentation edit scores of 80.48%.

The "Towards Real-Time Analysis of Broadcast Badminton Videos" paper (arXiv:2308.12199) is about providing a real-time analysis framework for broadcast videos, introducing an approach which generates heatmaps of player movements, calculates distance run by players and the average speed of player movements, performing point segmentation, player detection and homography based techniques.

The specific F1, precision, and recall metrics you mentioned don't appear in the search results provided. If you need those exact metrics, you may want to consult the full PDF of the paper directly.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

11. assistant (2026-06-06T17:33:15.342Z)

The exact metrics didn't surface in search. Let me read the local PDF that was saved, and also try the HTML version of the paper to get the exact numbers and the framing of the CNN task.

12. user (2026-06-06T17:33:18.515Z)

maxContentLength size of 10485760 exceeded

13. user (2026-06-06T17:33:23.441Z)

Exit code 2

/usr/bin/bash: eval: line 1: unexpected EOF while looking for matching `''

14. user (2026-06-06T17:33:26.271Z)

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```

15. assistant (2026-06-06T17:33:39.613Z)

The 2308.12199 PDF was saved locally (the largest one). Let me extract its text directly to find the exact Point Segmentation metrics and framing.

16. user (2026-06-06T17:33:41.666Z)

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=== webfetch-1780767185534-0drige.pdf ===
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/c/Users/avidu/AppData/Local/Programs/Python/Python313/python
/c/Users/avidu/AppData/Local/Microsoft/WindowsApps/python3
```

17. assistant (2026-06-06T17:33:44.186Z)

`pdftotext` is available. Let me extract text from the 2308.12199 PDF (the ~3.79MB one corresponds to the first webfetch of that paper).

18. user (2026-06-06T17:33:49.198Z)

<persisted-output>

Output too large (34.5KB). Full output saved to: C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\42211f80-2169-4025-ad9d-99c1aa29ff70\tool-results\b06zdoxam.txt

Preview (first 2KB):

webfetch-1780766966638-iygw3b.pdf

sf1638.cs05@nctu.edu.tw

Generally, there are several available datasets with frame-based or event-based records for various sports; for instance, Deli?ge et al. [9] proposed SoccerNet-v2 for soccer understanding and broadcast production via annotated soccer match videos, and Yue et al. [49] released SportsVU, which is a tracking dataset for games recorded from several seasons of NBA. However, in turn-based sports, there are only a few public datasets with tournament-level records [2, 30],

Although existing automatic detecting tools, e.g., [6], can be adopted to extract spatial-temporal stroke features from badminton videos, the automation introduces a constraint on the variability in the dataset since corresponding structured source data (i.e., the compositions of stroke, rally, set, and match) must be available to analyze the similarities and discrepancies between players within a match as well as across matches. Furthermore, providing highquality annotation requires serious efforts from domain experts since not only the number of videos is rapidly increasing but it is also hard for non-domain experts to distinguish the tactical metadata of a stroke. Therefore, one of the key challenges is to implement a labeling tool that can provide a straightforward process for labeling from each stroke to each match to reduce the difficulty of annotators reviewing videos repeatedly.

ShuttleSet contains 36,492 strokes, 3,685 rallies, and 104 sets from 44 matches between 2018 and 2021 played by 27 high-ranking men's and women's players. The details of annotation matches and the corresponding players are listed in Appendix A.1. Each match consists of 2 or 3 sets (best of 3), each of which is comprised of the metadata of strokes to describe the microscopic process from stroke to rally. The format of each stroke can be divided into four categories, which mainly expands from the BLSR design format [41]:

...

</persisted-output>

19. assistant (2026-06-06T17:33:52.254Z)

The first PDF appears to be a different paper (ShuttleSet-related, nctu.edu.tw). The webfetch caching may have returned a different doc. Let me directly download the correct 2308.12199 PDF and extract its Point Segmentation section.

20. user (2026-06-06T17:33:55.150Z)

```
-rw-r--r-- 1 avidu 197609 11475365 Jun  6 23:03 p2308.pdf
```

```
---EXTRACTED---
```

16:Abstract--Analysis of player movements is a crucial subset of sports analysis. Existing player movement analysis methods use recorded videos after the match is over. In this work, we propose an end-to-end framework for player movement analysis for badminton matches on live broadcast match videos. We only use the visual inputs from the match and, unlike other approaches which use multi-modal sensor data, our approach uses only visual cues. We propose a method to calculate the on-court distance covered by both the players from the video feed of a live broadcast badminton match. To perform this analysis, we focus on the gameplay by removing replays and other redundant parts of the broadcast match. We then perform player tracking to identify and track the movements of both players in each frame. Finally, we calculate the distance covered by each player and the average speed with which they move on the court. We

further show a heatmap of the areas covered by the player on the court which is useful for analyzing the gameplay of the player.

19:Sports analysis is one of the most challenging tasks in computer vision and several different approaches to sports analysis have been proposed. Sport analyses include match summarization [1], event prediction [2], ball tracking [3], structured analysis of the game [4] etc. This kind of analysis has proven to be useful in areas such as training and coaching,¹ and has also been able to provide help to the referee during the game. Apart from the types of analysis mentioned above, research topics like analysis and prediction of how groups of players move [5], automatic identification of key stages [6] in a game for gathering statistics or automating the control of broadcast cameras [7] etc. have also been explored. Game statistics are important to both viewers as well as the players and their coaches. For viewers, having access to accurate statistics improves viewing experience and helps them understand the more technical aspects of the sport like complex strategies and playing styles. It also aids the players and their coaches to formulate better game strategies and improve performance. Analysing an opponent's gameplay also helps players to adapt their style of play to suit the particular opponent. In badminton, games statistics include number of

22:Fig. 1: An overview of the distance travelled by the player from the start of a set. Player in blue bounding box refers to the "top-player" and player in yellow bounding box refers to the "bottom-player". Dot at the center-bottom of the player's bounding box represents their current location. The "Rally Duration" is the time length of a particular rally. Four such rallies are depicted for reference. (best viewed in color)

26:

2. Many of the existing analysis methods are constrained by these challenges. Our main focus in this work is to provide real-time analysis framework for the broadcast videos which is expected to generalize to almost all the badminton games captured from broadcast camera view.

27:In this work, we introduce an approach which generates heatmaps of player movements, calculates distance run by players and the average speed of player movements for the game of badminton (figure 1). We perform point segmentation, player detection and homography based techniques to compute the above defined metrics.

32:4) We introduce two new datasets, one each for singles and doubles matches covering a wide range of broadcasting camera angles, court and lighting conditions, and featuring several different players. We also introduce a set of 25 rallies badminton match video of our own recording having the ground truth data for the distance covered by the players in each rally.

48:

Fig. 2: Our dataset contains various broadcast camera positions, illuminations settings, different backgrounds and court colors which helps generalize the framework for any kind of badminton match scenario. It should be observed that having audiences in the background may affect the player localization method if not generalized well.

62:TABLE I: A comparison of our datasets and the Badminton Olympic Dataset [4]. We show that our dataset is more diverse as we cover more tournaments, capturing multiple broadcast camera angles and a variety of illumination settings in the courts and background.

67:A. Point Segmentation Annotation

68:Broadcast badminton matches are divided into gameplays and replays. We define a "rally" frame as a frame where the actual gameplay is happening and a "non-rally" frame as a frame where the replay (highlights) has been shown, or players are resting, etc. (or in other words whether the gameplay is not happening). Generally, non-rally frames contain redundant information about the game and do not contribute much in the analysis part. Therefore, we first start classifying between rally and non-rally frames. We call this classification method as Point Segmentation. This allows us to keep all the rally frames, group them into rallies according to the points scored by the players and discard the non-rally frames. To make the dataset usable for point segmentation, we annotate every frame

70:of a match as a rally or non-rally frame using the ELAN video annotation tool ⁴. We annotate 23156 frames in total, where 15147 were non-rally frames and 8009 were rally frames.

72:We annotate the dataset to get the bounding box for each player using the LabelImg⁵ graphical image annotation tool. We define the player's position in the court with respect to the broadcast camera point of view where one player plays from the near side, and other player plays from the far side of the camera, and we define them as bottom and top player respectively (see figure 1 for more details). We manually annotate players' bounding boxes with two classes "PlayerTop" and "PlayerBottom" where "PlayerTop" corresponds to the player on the far side of the court and while "PlayerBottom" corresponds to the player on the near side of the court with respect to the viewpoint of the camera. We randomly pick 150 rally frames from each match and annotated them for player bounding boxes for both the players. Thus, we end up with 3000 (150 frames ? 2 players ? 10 matches) annotated frames in total. We use this annotation to learn player bounding boxes for the rest of the frames. We use the same annotation style for the doubles matches as well. As shown in figure 4, occasionally, players are occluded or blurred due

to the fast-paced nature of the game. Such occlusions are even more prevalent in the doubles matches where there are four players on court as opposed to the two players in a singles match.

Fig. 3: We propose a real-time end-to-end pipeline to analyze live broadcast match videos. Our pipeline consists of point segmentation, player localization, pictorial point summary and top-view space conversion to compute the distance and overall analysis.

79:which were accurate up to 1 meter, measured the distance covered by each player during each rally. We recorded 25 rallies and obtained the distance values for each player for all of them. All the videos were recorded at approximately the same broadcast angle as that of the singles and doubles datasets using a smartphone camera.

81:As shown in figure 3, our pipeline is largely based on the approach proposed by Ghosh et al [4]. It comprises of 4 steps: (1) Point Segmentation (2) Player Localization (3) Top View Space Conversion and (4) Analysis. We optimize each of these steps independently to make them more robust while simultaneously reducing latency. The improved pipeline can be used for real-time analysis of videos.

82:A. Point Segmentation

83:The very first step of our pipeline is to keep the rally frames and discard the non-rally frames (which are redundant for our analysis) from the given match video. Rally frames are usually always shot from the broadcast camera view whereas the non-rally frames can include close-ups of players, additional graphics (score cards) and unusual camera angles. These differences help distinguish these frames. Unlike Ghosh et al [4] who use HOG + SVM, we use a convolutional neural network based classifier to classify rally and non-rally frames. We finetune the ResNet-18 [24] model (pre-trained on ImageNet [25] dataset) by changing the last layer of this network to label the frame as rally frame or non-rally frame. As shown in table II, our CNN based method is both more accurate and also exhibits lower latency compared to HOG + SVM.

84:At test time, we classify the frames retrieved from the match video and number the rally frames in a sequential manner.

86:We assume that a rally is at least 3 seconds long and there is a minimum of 3 seconds gap between two rallies. This allows us to group frames of the match into individual rallies by separating sets of continuous rally frames with at least 3 seconds of non-rally frames in between them. Thus Point Segmentation gives us a set of rallies from the match. We number each rally sequentially as rally-1, rally-2 etc.

88:Once we obtain individual rallies, we need to find player location in each frame of each rally. A YOLOv3 [26] network is finetuned for these two classes. In order to overcome the constraint of only having 3000 annotated samples for player localization, we use transfer learning to train the YOLOv3 network. We first pre-train the YOLOv3 network on PASCALVOC dataset [27] which has 20 classes including the "Person" class. Further, we finetune the YOLOv3 network on our own dataset to achieve better accuracy. We used the same method for both the single's and double's matches to get the location of the player. However, as can be seen in figure 4, player localization is a very difficult problem to solve in doubles matches.

89:Using a convolutional network for point segmentation and YOLOv3 for player localization helps us build a fast and robust pipeline that performs better than [4] both in terms of accuracy of predictions and latency. The improved accuracy as well as inference time make it possible to use our pipeline for a real-time implementation. Please refer to table II for more details.

98:F1-Score

99:95.44 97.23

101:Precision

102:97.83 98.06

104:Recall

105:91.02 92.34

110:Fig. 5: Conversion of player locations from camera coordinates to top view co-ordinates. The top-view space provides the player's position and footwork information around the court for a particular gameplay. This view can also be seen as heatmap of the players for that particular rally. (Best viewed in color)

112:(a) Point Segmentation Method

125:Rally Latency (Seconds)

133:side of the camera. In order to find out the distance or other kind of analysis, we need to switch the view of player positions from broadcast camera view to another view where the displacement of the player is not scaled by the near side or far side of the player position. We choose the top view of the badminton court where we map the player locations from broadcast camera view. So, using homography techniques, we map the 2-dimensional broadcast view camera coordinates to top view court coordinates. By doing this, we get an equivalent linear top view for the player locations. To perform homography calculation, we find the court lines from the video frame using Hough line finding method [28] and get the position of court corners. We

compute a homography matrix between these four court corners and top view four court corners to get a mapping from camera coordinates to top view space coordinates. Finally, we map the player locations from broadcast camera view to top view. Please refer to figure 5 for output details.

135:Player Distance Calculation: To compute the distance covered by the player in a particular rally, we utilize our player tracks in top view space. Now the player locations are in 2dimensional space and we take the "center bottom" point of the player bounding box as a proxy for the player's current location.

136:According to the standard badminton rules⁶, the length and width of the badminton court is 13.41 meters and 6.1 meters respectively. From this information, we know that from the top left corner of the court to the top right corner of the court, the distance of pixels is 6.1 meters and similarly, from the top left corner of the court to the bottom left corner of the court the distance is 13.41 meters. We use this information to calculate the distance of the player covered in meters. We calculate the euclidean distance between the locations of the player in successive frames for both players. Now we have the pixel-based distance between two successive frames. We use unitary method to scale this pixel-based distance into meters to get the actual distance values. We add all the distances (meters) successively for the whole rally to calculate the distance for each player in that particular rally. The distance calculation of

139:

a rally for one player is formulated as:

165:where n is the number of frames in the rally, x_i and y_i are the co-ordinates of the player in the top-view space at i th frame, D_h is the pixel-based distance from top-left corner to the top-right corner (horizontal distance) and D_v is the pixel-based distance from top-left corner to the bottom-left corner (vertical distance). For the game of double's, we have two players on each side of the court i.e. two "PlayerTop" (pt) and two "PlayerBottom" (pb). Therefore, for the first frame, we initialize pt1 and pt2 for the two top players and pb1 and pb2 for the two bottom players respectively. In subsequent frames, we use the player's previous position to track them individually. Again, we use the method same as singles to calculate the distance.

167:Player Average Speed Calculation: To calculate the average speed achieved by each of the players in a particular rally, we use the distance covered by the players. As we know the number of frames in a particular rally, we find out the total time elapsed (25 frames per second) and divide the player distance covered by the total time.

171:To evaluate the accuracy of our proposed pipeline in terms of the distance covered by the players in a particular rally, we used our recorded badminton match video which has ground truth data of 25 rallies for the distance covered by the players. We use our proposed pipeline to obtain predicted distance values for each rally and compare them to the ground truth values. We used the same evaluation metric as [29], which uses paired t-tests to calculate differences between the ground truth distances and predicted distances. We report the mean, standard deviation, minimum, and maximum error between the ground truth and predicted values for both "Player Top" and "Player Bottom". We also report the t-stats value and degree of freedom of the paired t-test calculation between the ground truth and predicted values. Please refer to table III for the report.

187:The proposed pipeline was successfully used in a real-world setting to analyze live broadcast matches in real-time during the Premier Badminton League 2019. As reproducing the exact same scenario is not feasible, we use a simulation of the realtime scenario to evaluate the performance (with respect to latency) of our model. Generally, as live matches are streamed, the camera recording the match captures 25 frames per second and thus the match video file size increases with time. In order to simulate the real-time scenario, we fed the pre-recorded match videos to our proposed pipeline at precisely 25 frames per second, sequentially. Thus our models did not have access to all the frames at the same time. We ran this simulation on 61 rallies (totaling up to 55 minutes) and timed our models. Figure 6 shows the results of this simulation.

189:We use the live feed of the broadcast video match from Premier Badminton League - 2019 (courtesy of Star Sports India⁷). We used frames from the live video in batches of 20 for our experiments. We attempt to extract the analysis from the data we got using the methods mentioned above. We have segmented the frames into either rally or non-rally frame using point segmentation method which in the end, gives us rally directories as soon as the rally is over in the live match, where each directory contains rally frames for that specific rally. Then we pass the rally frames to the player localization method to get the player bounding boxes. We perform homography to change the space from broadcast camera view to top-view. We perform each analysis on frame level for the accessed file.

192:

On an average, one rally is 10 to 12 seconds long and it contains 250 to 300 rally frames. For a real time implementation, it is not feasible to process these many frames in one go. Hence, to solve this problem for this scenario, we process every 3rd rally frame for player localization method. As there are 25 frames per second in the broadcast match video, we assume that there is no significant change in player location if we take every third rally frame. After getting the

rally directories from the above point segmentation method we pass the rally directories to the player localization method to get the player location for each of the frames in a parallel manner. Using this rally-level parallelism for analysis helped us to achieve the goal in the real-time scenario.

197:Another problem that hinders research in the area of sports analysis, and more specifically, badminton analysis, is the lack of labelled datasets. We introduce 3 different datasets to help combat this problem. Two of our datasets (singles and doubles) consist of Badminton World Federation match videos and feature professional badminton players and official courts and recording equipment. Both these datasets are annotated for Point Segmentation and Player Localization. The third dataset consists of amateur players recorded by using a smartphone camera from the approximately same broadcast angle as the other datasets. This dataset also consists of ground truth data for the distance covered by the players in each rally (which was obtained from the distance trackers worn by the players during gameplay). We hope that access to these datasets helps spur further research into real-time analysis of badminton videos.

21. assistant (2026-06-06T17:34:14.181Z)

I now have the full primary-source text. Let me assess this carefully against the claim.

****What the paper actually says (verbatim, confirmed):****

- Task: "We define a 'rally' frame as a frame where the actual gameplay is happening and a 'non-rally' frame as a frame where the replay (highlights) has been shown, or players are resting, etc." (line 68)
- Method: "we use a convolutional neural network based classifier to classify rally and non-rally frames. We finetune the ResNet-18 model (pre-trained on ImageNet)... to label the frame as rally frame or non-rally frame." (line 83)
- "Rally frames are usually always shot from the broadcast camera view whereas the non-rally frames can include close-ups of players, additional graphics (score cards) and unusual camera angles." (line 83) ? matches the supporting quote exactly.
- Metrics (Table II, lines 99/102/105): the two columns are HOG+SVM [Ghosh et al.] vs the CNN. F1: 95.44 / ****97.23****; Precision: 97.83 / ****98.06****; Recall: 91.02 / ****92.34****. The CNN column matches the claim exactly.
- Dataset: 23,156 frames total (15,147 non-rally, 8,009 rally) (line 70).
- No shuttle tracking ? the pipeline is Point Segmentation ? YOLOv3 player localization ? homography ? distance. The CNN classifier uses no shuttle trajectory. Confirmed.

So the metrics, architecture, "no shuttle tracking," and the broadcast-vs-close-up framing are all faithfully reproduced. The claim is well-supported by the primary source on its literal terms.

****The critical nuance to test (the refutation angle):**** The claim says "point/rally segmentation can be done by a per-frame CNN ... achieving F1 97.23." But these metrics are ****per-frame binary classification accuracy**** (rally vs non-rally frame), NOT temporal-boundary/IoU segmentation accuracy. The paper's actual rally grouping relies on a heuristic ("a rally is at least 3 seconds long and there is a minimum of 3 seconds gap" ? line 86), and the 97.23 F1 is the frame-classifier's F1, not a rally-boundary F1. Let me verify whether the claim conflates these, and check the key context dependency: this works because of broadcast-specific visual cues (replays/close-ups/scorecards). Let me confirm there's no separate boundary metric and check generalizability concerns.

22. user (2026-06-06T17:34:14.476Z)

=== Table II context ===

93-

Fig. 4: Output of player tracking method. It should be noted that this method works well, even when there are levels of occlusion and irrespective of player poses.

94-

95-Method

96-HOG+SVM CNN

97-

98-F1-Score

99:95.44 97.23

100-

101-Precision

=== Any IoU / boundary / temporal segmentation metric? ===

24:While considerable research has been published regarding badminton analysis, the one common limitation of all existing methods is that they are applied post-factum on previously recorded match videos. This prevents the use of these methods to show statistics during a live broadcast match. The ability to compute these statistics in real-time from a live broadcast match not only makes the game more exciting for viewers but also allows the players to dynamically modify their strategies mid-game. However, broadcast videos are captured from multiple viewpoints for the best viewing experience and their 'unstructured' nature along with the rapid movement and complex human pose and motion makes real-time analysis a complex task. Moreover, broadcast delay for a typical badminton match is less than 15 seconds, usually 7 seconds

29:1) We propose an end-to-end framework (figure 3) to analyze live broadcast match videos, the components of which can be trained using pre-recorded badminton matches. Unlike previous approaches, our method uses visual cues only and does not rely on any hardware setup.

39:Our work is inspired by the work of [4] that proposes an end-to-end framework to automatically annotate badminton broadcast videos. Their work identifies various understandable metrics that can be computed for analyzing badminton matches which in turn help qualitative understanding of badminton games. We build on top of [4], and extend the work further by (a) covering wider match situations/tournaments/games (b) adapting the solution to real-time deployment and (c) a successful field trial in PBL 2019 with commentators and broadcasters appreciating the utility.

48:

Fig. 2: Our dataset contains various broadcast camera positions, illuminations settings, different backgrounds and court colors which helps generalize the framework for any kind of badminton match scenario. It should be observed that having audiences in the background may affect the player localization method if not generalized well.

88:Once we obtain individual rallies, we need to find player location in each frame of each rally. A YOLOv3 [26] network is finetuned for these two classes. In order to overcome the constraint of only having 3000 annotated samples for player localization, we use transfer learning to train the YOLOv3 network. We first pre-train the YOLOv3 network on PASCALVOC dataset [27] which has 20 classes including the "Person" class. Further, we finetune the YOLOv3 network on our own dataset to achieve better accuracy. We used the same method for both the single's and double's matches to get the location of the player. However, as can be seen in figure 4, player localization is a very difficult problem to solve in doubles matches.

110:Fig. 5: Conversion of player locations from camera coordinates to top view co-ordinates. The top-view space provides the player's position and footwork information around the court for a particular gameplay. This view can also be seen as heatmap of the players for that particular rally. (Best viewed in color)

165:where n is the number of frames in the rally, x_i and y_i are the co-ordinates of the player in the top-view space at i th frame, D_h is the pixel-based distance from top-left corner to the top-right corner (horizontal distance) and D_v is the pixel-based distance from top-left corner to the bottom-left corner (vertical distance). For the game of double's, we have two players on each side of the court i.e. two "PlayerTop" (pt) and two "PlayerBottom" (pb). Therefore, for the first frame, we initialize pt1 and pt2 for the two top players and pb1 and pb2 for the two bottom players respectively. In subsequent frames, we use the player's previous position to track them individually. Again, we use the method same as singles to calculate the distance. (none above = no boundary metric)

=== License ===

75:<https://tla.mpi.nl/tools/tla-tools/elan/> 5<https://github.com/tzutalin/labelImg>

23. assistant (2026-06-06T17:34:34.646Z)

This confirms my analysis. Let me consolidate the assessment.

****Verification results:****

1. ****Claim supported by quote/source?**** YES, on its literal terms. Every number (F1 97.23, P 98.06, R 92.34), the ResNet-18 CNN classifier, the "broadcast camera vs close-ups/replays" framing, and "no shuttle tracking" are reproduced faithfully from the primary source (arXiv 2308.12199, "Towards Real-Time Analysis of Broadcast Badminton Videos," Table II, Section IV-A). The supporting quote is verbatim.

2. ****Source quality:**** Primary source, peer-style arXiv paper from IIIT/CVIT lineage, field-trialed at PBL 2019. Adequate for the claim's strength.

3. **Contradicting evidence:** None found that disputes the numbers. The related precursor (Ghosh et al. "Towards Structured Analysis") reports HOG+SVM at 95.44 F1, consistent with the paper's own comparison column.

However ? important qualifications that the claim's framing glosses over (refutation considerations):

- **It is per-frame classification accuracy, not temporal-boundary segmentation.** The 97.23 F1 measures whether each frame is correctly labeled rally/non-rally ? there is NO IoU/boundary/segment-level metric in the paper (confirmed: no IoU/edit-score/tIoU anywhere). Actual rally *boundaries* come from a hand-tuned heuristic ("a rally is at least 3 seconds long, min 3 seconds gap"). So "rally segmentation ... achieving F1 97.23" slightly overstates: 97.23 is frame-classification F1, not rally start/end localization F1 (which is exactly the IoU-0.5 metric our project cares about). These are not the same thing.

- **It exploits BROADCAST-specific cues** (replays, score-card graphics, close-ups, camera cuts). The whole signal is "broadcast camera view vs everything else." This is a critical context dependency: it does NOT generalize to single-fixed-camera amateur/self-recorded footage where there are no replays, no director cuts, no close-ups ? the video is one continuous broadcast-like shot, so a "broadcast vs non-broadcast" classifier has nothing to discriminate. The claim states the method as if it's general rally segmentation; it is really "cut out the non-gameplay portions of a TV broadcast."

These are scope qualifications, not refutations. The claim as literally worded ("Badminton point/rally segmentation can be done by a per-frame CNN ... achieving F1 97.23 ... without shuttle tracking") is accurately quoting a real, primary-source result. The conflation of frame-classification F1 with temporal-segmentation F1 is a mild overreach in wording but the underlying factual claim (CNN classifier, those metrics, no shuttle tracking) is true and well-sourced. Under the rubric, this is not "unsupported / contradicted / low-quality / outdated / marketing." I will not refute.

24. user (2026-06-06T17:34:43.910Z)

Structured output provided successfully

25. assistant (2026-06-06T17:34:49.570Z)

Verification complete. The claim is NOT refuted ? every factual element (ResNet-18 CNN classifier, F1 97.23 / Precision 98.06 / Recall 92.34, "broadcast camera vs close-ups/replays" framing, no shuttle tracking, 23,156-frame dataset) is faithfully reproduced from the primary source (arXiv 2308.12199, Table II and Section IV-A), which I downloaded and extracted directly.

Two scope qualifications worth carrying into the research report (they qualify rather than refute the claim): the 97.23 F1 is per-frame rally/non-rally classification accuracy, not temporal start/end (IoU) segmentation ? the paper has no boundary/IoU metric and uses a hand-tuned heuristic for the actual grouping; and the discriminative signal is broadcast-specific (replays, score-card graphics, director cuts, close-ups), so it will not transfer to single-fixed-camera amateur/self-recorded footage.